

|                                                 |                                                          |
|-------------------------------------------------|----------------------------------------------------------|
| Program : <b>Diploma in Printing Technology</b> |                                                          |
| Course Code : <b>6101B</b>                      | Course Title: <b>Green Printing and Waste Management</b> |
| Semester : <b>6</b>                             | Credits: <b>4</b>                                        |
| Course Category: <b>Program Elective</b>        |                                                          |
| Periods per week: <b>4 (L:4, T:0, P:0)</b>      | Periods per semester: <b>60</b>                          |

### Course Objectives:

- To impart the knowledge on the need and importance of employing ecofriendly process in Print production.
- To guide the student to understand the nature of waste generated in various areas of Printing industry and various control measures for waste management during production.

### Course Prerequisites:

| Topic                                                   | Course code | Course Title                       | Semester |
|---------------------------------------------------------|-------------|------------------------------------|----------|
| Offset printing process.                                |             | Mechanics of Printing Machines-I   | 3        |
| Paper, Ink, Plate, Chemicals used in printing industry. |             | Materials for Printing & Packaging | 3        |
| Flexography, Gravure, Screen printing process.          |             | Mechanics of Printing Machines-II  | 4        |
| Digital Imaging techniques.                             |             | Digital printing                   | 4        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                             | Duration (Hours) | Cognitive Level |
|-----------------|-----------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Explain about various aspects of eco friendly - raw materials used in Print production. | 14               | Understanding   |
| CO2             | Describe the types of waste produced in various printing industries.                    | 12               | Understanding   |

|     |                                                                              |    |               |
|-----|------------------------------------------------------------------------------|----|---------------|
| CO3 | Demonstrate the automation techniques implemented in print media production. | 18 | Understanding |
| CO4 | Describe the waste management techniques.                                    | 14 | Understanding |
|     | Series Test                                                                  | 2  |               |

### CO-PO Mapping:

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     |     |     |     |     |
| CO2             |     |     |     |     | 2   |     |     |
| CO3             | 2   |     |     |     |     |     |     |
| CO4             |     |     |     |     | 3   |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

### Course Outline:

| Module Outcomes                                                                                                                                                                                                                                                                                                                                                                                                         | Description                                                                                   | Duration (Hours) | Cognitive Level |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Explain about various aspects of eco friendly -raw materials used in Print production.</b> |                  |                 |
| M1.01                                                                                                                                                                                                                                                                                                                                                                                                                   | Eco friendly production.                                                                      | 5                | Understanding   |
| M1.02                                                                                                                                                                                                                                                                                                                                                                                                                   | Eco friendly printing Ink & Paper.                                                            | 3                | Understanding   |
| M1.03                                                                                                                                                                                                                                                                                                                                                                                                                   | Eco friendly offset plates.                                                                   | 3                | Understanding   |
| M1.04                                                                                                                                                                                                                                                                                                                                                                                                                   | Eco friendly chemicals.                                                                       | 3                | Understanding   |
| <b>Contents:</b><br>Definition of green printing. Eco friendly printing paper - types-recycled paper, wood free paper, etc. Eco friendly inks - types - vegetable based ink, bio inks etc. Ink recycling. Eco friendly offset plates - low chemical, chemical free, etc. recycling of plates. Eco friendly chemicals - IPA substitutes, bio chemicals etc. Eco friendly production - waste reduction, waste management. |                                                                                               |                  |                 |
| CO2                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Describe the types of waste produced in various printing industries.</b>                   |                  |                 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                     |   |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---|---------------|
| M2.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Types of waste produced in Printing industry.                                       | 5 | Understanding |
| M2.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Ways to reduce waste during production.                                             | 4 | Understanding |
| M2.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Make ready time.                                                                    | 3 | Understanding |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Series Test - I                                                                     | 1 |               |
| <b>Contents:</b><br>Types of waste produced in - screen printing, offset printing, flexographic printing, gravure printing, digital printing. Ways to reduce waste during production in - screen printing, offset printing, flexographic printing, gravure printing, and digital printing. Optimization. Make ready time. Make ready time comparison between manual and automated systems. Ways to reduce make ready time in sheet fed and web fed machines.                                                                         |                                                                                     |   |               |
| <b>CO3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Demonstrate the automation techniques implemented in print media production.</b> |   |               |
| M3.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Advanced presses.                                                                   | 8 | Understanding |
| M3.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Automation in print media production.                                               | 7 | Understanding |
| M3.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Make ready for Print Production.                                                    | 3 | Understanding |
| <b>Contents:</b><br>Advanced presses - press remote control console, Ink feed preset systems, color measurement and control system, Register measurement and control, inline print quality measurement and control systems, The State of the Art in Remote Control, Measurement and Control Systems. Automation in print media production -wash up procedure, plate changing, sheet size and paper travel adjustments, Ink Feed Presetting in the Printing Unit, Ink and Dampening Solution Supply. Make ready for Print Production. |                                                                                     |   |               |
| <b>CO4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Describe the waste management techniques.</b>                                    |   |               |
| M4.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Types of waste.                                                                     | 5 | Understanding |
| M4.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Recycling of print materials.                                                       | 5 | Understanding |
| M4.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Printing waste management.                                                          | 4 | Understanding |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Series Test II                                                                      | 1 |               |
| <b>Contents:</b><br>Types of waste - solid, water, air. Reduce - Reuse - Recycle. Recycling of print materials and consumables. Printing waste management - Solid waste management. Solvent management. Ink management. Hazardous waste management. Air emission management. Noise management. Water and energy consumption.                                                                                                                                                                                                         |                                                                                     |   |               |

**Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                                                                    |
|------------|---------------------------------------------------------------------------------------------|
| T1         | P. Green, <b>Understanding Digital Colour</b> , GATF, 1995                                  |
| R1         | A. Sharma, <b>Understanding Colour Management</b> , Thompson Dalmar Learning                |
| R2         | Michael Bernard, John Peacock, <b>Handbook of printing and production.</b>                  |
| R3         | Helmut Kipphan, <b>Hand book of print media</b> , Springer Science & Business. Media        |
| R4         | HL Apfelberg and MJ Apfelberg, <b>Implementing quality management in graphic arts GATF.</b> |

**Online Resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                                                                                                                             |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://p2infohouse.org/ref/42/41802.pdf">https://p2infohouse.org/ref/42/41802.pdf</a>                                                                                 |
| 2            | <a href="https://cdn.aigroup.com.au/Environment/10_Printing_Waste_Reduction_Factsheet.pdf">https://cdn.aigroup.com.au/Environment/10_Printing_Waste_Reduction_Factsheet.pdf</a> |